Complex Numbers: Algebra

Questions 1–48

Question 1. Write the following as imaginary numbers:

a) $\sqrt{-9}$
b) $\sqrt{-2}$
c) $2\sqrt{-4}$
d) $-3\sqrt{-2}$

Question 2. Simplify $\sqrt{-a} \times \sqrt{-b}$ where a and b are real positive numbers.

Question 3. Write down the simplified form of each of the following:

a) i^2
b) i^3
c) i^4
d) i^5
e) i^{40}
f) i^{41}
g) i^{99}

Question 4. Write down the simplified form of each of the following, where n is a positive integer:

a) i^{4n}
b) i^{4n+1}
c) i^{4n+2}
d) i^{4n+3}
e) i^{4n+17}
f) i^{4n+26}

Question 5. Determine the sum of the expression

$$i + i^2 + i^3 + \cdots + i^{1000}$$

Fully justify your answer.

Question 6. Simplify

a) $(1-i)^2$
b) $(1-i)^3$
c) $(1-i)^4$
d) $(1+i)^2$
e) $(1+i)^3$
f) $(1+i)^4$

Question 7. Without using a calculator, simplify each of the following:

a) $\sqrt{-12} \times \sqrt{-27}$

b) $(2i)^3$

c) $i^2(i+3)$

d) $i + 4i^5$

e) $(i - 3i^3)(1 - 2i^4)$

f) $(2i^6 - i^9)^2$

Question 8. Evaluate

$$\left(i^{12} + \left(\frac{1}{i}\right)^{18}\right)^3$$

Question 9. Evaluate

$$\left(i^{30} - \left(\frac{1}{i}\right)^{12}\right)^4$$

Question 10. Given that $z_1 = 2 - 3i$, $z_2 = 1 + i$ and $z_3 = 5 - 2i$ find each of the following:

a) $z_1 + z_2$
b) $z_1 + z_3$
c) $z_3 - z_1$
d) $2z_2 - z_1$
e) $5z_1 + 3z_2$

Question 11. Show that $(2+3i)^2 = -5+12i$

Question 12. Show that $(-1+5i)^3 = 74-110i$

Question 13. Given x and y are real numbers, find the real and the imaginary parts of each of the following:

a) $z = 2 + (3x - iy)$

b) $z = (x - 2i) - (2x + 3iy)$

c) $z = 5(x - iy) + 2(x + iy)$

Question 14. Find the value(s) of real numbers x and y that satisfy each of the following equations:

a) $5x + 2y + (3x + 4y)i = 25 + 29i$

b) $2x + 8y + (x - 4y)i = -20 + 14i$

c) $(-3x + 2ix) + (2y + 7iy) = -14 - 74i$

d) $x + 3x^2 + (9x - y)i = 2 + i$

Question 15. Find the value of real number a such that

$$(3+7i)(a+2i)$$

is purely imaginary.

Question 16. Given that

$$(2+i)(a+bi)$$

is purely imaginary, simplify the expression giving your answer only in terms of a

Question 17. Given that

$$(a - bi)(a + 3bi) = 52 + 16i$$

where a and b are real integers, find the possible values of a and b

Question 18. Write each of the following in the form $z^2 + pz + q$ where p and q are complex numbers:

a) $(z + 1 - 2i)^2$

b) $(z - 2 + 3i)^2$

c) $(z + 1 + 2i)(z - 2 + 3i)$

Question 19.

a) Simplify $(7+2i)(3+8i)$

b) Simplify $(7-2i)(3-8i)$

c) It is given that $5^2+62^2=3869$. Using this and your answer to part a) and part b), find the prime factorisation of 3869

Question 20. Determine the possible values of z for each of the following:

a) $z^2=5+12i$

b) $z^2=21-20i$

c) $z^2=\frac{3}{16}-\frac{1}{4}i$

Question 21. Find, giving your answer in exact form:

a) $\sqrt{4+3i}$

b) $\sqrt{-12+16i}$

c) $\sqrt{2+4\sqrt{2}i}$

Question 22. Solve the simultaneous equations below to find complex numbers w and z:

$$2iz + 4w = -6 + 8i$$
$$-3z + 5wi = 7 - i$$

Question 23. For each of the following, find the possible value(s) of the unknown real value(s) a:

a) $\text{Re}[(1+ai)^3] = 0$
b) $\text{Im}[(3+ai)^3] = 0$
c) $\text{Im}[(4+ai)^4] = 0$
d) $\text{Re}[(2-ai)^4] = 0$
e) $\text{Im}[(1-ai)^5] = 0$

Complex Conjugates

Question 24. Write down the complex conjugate, z^* for each of the following:

a) $z = 2 + i$
b) $z = 3 - i$
c) $z = -6 - i$
d) $z = 1 + 5i$
e) $z = -4 - 2i$
f) $z = \frac{1}{2} + \frac{3}{4}i$
g) $z = \sqrt{5} - \sqrt{3}i$
h) $z = 2i$
i) $z = -10$

Question 25. Given the complex number $z = a + bi$, prove that each of the following are real numbers:

a) $z + z^*$

b) zz^*

Question 26. Given $w = 2 - i$ and $z = -5 + 3i$, find:

a) $2w - z$

b) $w^* + w$

c) $2w^* + z$

d) z^2

e) wzz^*

f) $w(z^*)^2$

g) $(w^*)^3$

Question 27. Given $z = x + iy$

a) Find the real and imaginary parts of $4iz + 3z^*$

b) Hence find the values of x and y such that $4iz + 3z^* = 5 - 3i$

Question 28. Find the complex number z if

a) $2z^* = 3 - 4i$

b) $3z^* + 1 = 8i$

c) $iz^* = -1 + i$

Question 29. Find z given

a) $z + 2z^* = 3 + 9i$

b) $2z - z^* = -4 - 5i$

c) $z + 5z^* = -i$

Question 30. Prove that

a) $(z^*)^2 = (z^2)^*$

b) $(z^*)^3 = (z^3)^*$

Question 31. Given the complex number $z = a + bi$, where a and b are real numbers, find

a) $z - z^*$
b) $(z - z^*)^2$
c) $(z - z^*)^3$
d) $(z - z^*)^4$
e) $(z - z^*)^{18}$

Question 32. Find the four solutions to the equation

$$(4z - z^*)^* = z^2$$

Question 33. The two complex numbers z and w satisfy

$$2z + 4z^* = 18 + 4i$$
$$5w - 3z^* = -3 + 2i$$

Find the value of w in the form $a + bi$

Question 34. A complex number z satisfies the equation,

$$z^2 + 3iz^* = 8$$

where $\mathrm{Re}(z) < 0$. Find z in the form $a + bi$ where a and b are real constants to be found.

Question 35. A complex number z satisfies the equation,

$$z^2 + 4z^* = -4 + 10i$$

where $\mathrm{Re}(z) > 0$. Find z in the form $a + bi$ where a and b are real constants to be found.

Rationalising the Denominator

Question 36. Without using a calculator, simplify each of the following:

a) $\frac{4}{i}$
b) $\frac{7}{1+i}$
c) $\frac{9}{5-2i}$
d) $\frac{8}{2+3i}$
e) $\frac{2+i}{4-i}$
f) $\frac{(1-i)^2}{1+i}$
g) $\frac{a+bi}{b-ai}$

Question 37. Given that $z = -1 + 2i$, express $z + \frac{5}{z}$ in the form $a + bi$ where a and b are real constants to be found.

Question 38. Ohm's law states that $V = IR$ where I is the current through the conductor (amps), V is the voltage measured across the conductor (volts), and R is the resistance of the conductor (ohms).

a) Given an electrical circuit has a current of $(3 - 2i)$ amps and a resistance of $(4 + i)$ ohms, find the voltage for this circuit.

b) Given the voltage on a circuit is $(6 - 2i)$ volts and a current of $(2 + 3i)$ is passed through the conductor, calculate the resistance of the conductor giving your answer in its simplified form.

Question 39. Find z given that

$$\frac{15-5i}{z} = 2-3i$$

Question 40. Evaluate

$$\frac{2}{3+i} + \frac{3}{2+i}$$

Question 41. Express

$$\frac{1}{\cos\theta - i\sin\theta}$$

in the form $a + bi$

Question 42. Given x and y are real, find the real and the imaginary parts of each of the following:

a) $z = x + iy + \frac{1}{x+iy}$
b) $z = x + iy - \frac{1}{x+iy}$
c) $z = x - iy + \frac{1}{x-iy}$
d) $z = x - iy - \frac{1}{x-iy}$

Question 43. Find the least positive integer of n for which

$$\left(\frac{1+i}{1-i}\right)^n = 1$$

Question 44. Find the values of integers a and b such that

$$\frac{a}{1-2i} + \frac{b}{3-2i} = \frac{-5+6i}{1+8i}$$

Question 45. Given $z = a + bi$, and that

$$zz^* = 17 \qquad \frac{z}{z^*} = \frac{15}{17} - \frac{8}{17}i$$

determine the possible values of a and b

Question 46. Given the complex number $z = a + bi$, find the real and imaginary parts of

$$\frac{z}{z-2}$$

Question 47. Find z given that

$$\frac{z}{3+4i} + \frac{z-1}{5i} = \frac{5}{3-4i}$$

Question 48. Given that

$$x - iy = \sqrt{\frac{a-bi}{c-di}}$$

where a, b, c and d are real numbers, show that

$$x^2 - y^2 = \frac{ac+bd}{c^2+d^2}$$